



Data Science Internship at Codveda Technologies

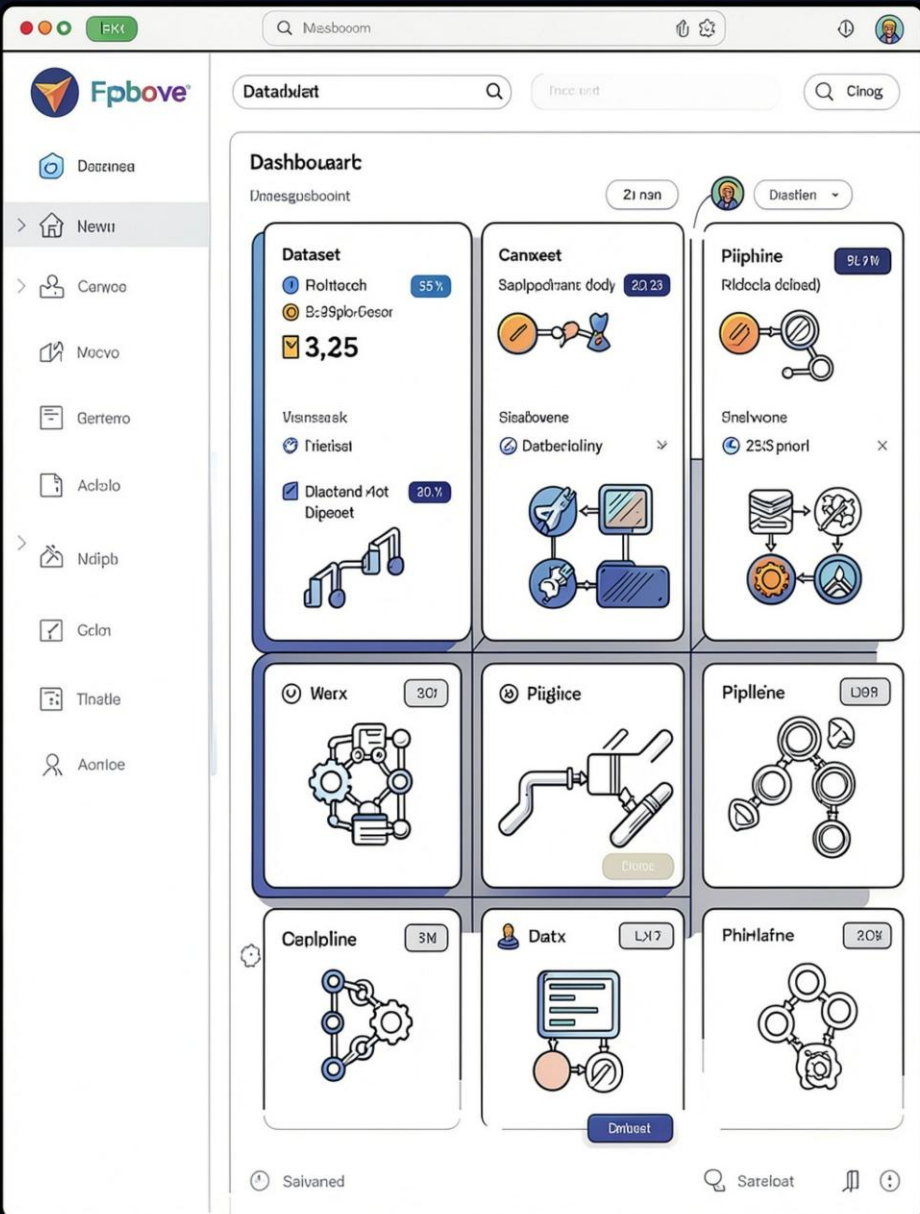
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Course - MCA

University - IILM University, Greater Noida

Internship Status - Ongoing



Presentation Overview

Context

Internship scope,
objectives,
datasets

Workflow

Data pipeline →
Modeling →
Evaluation

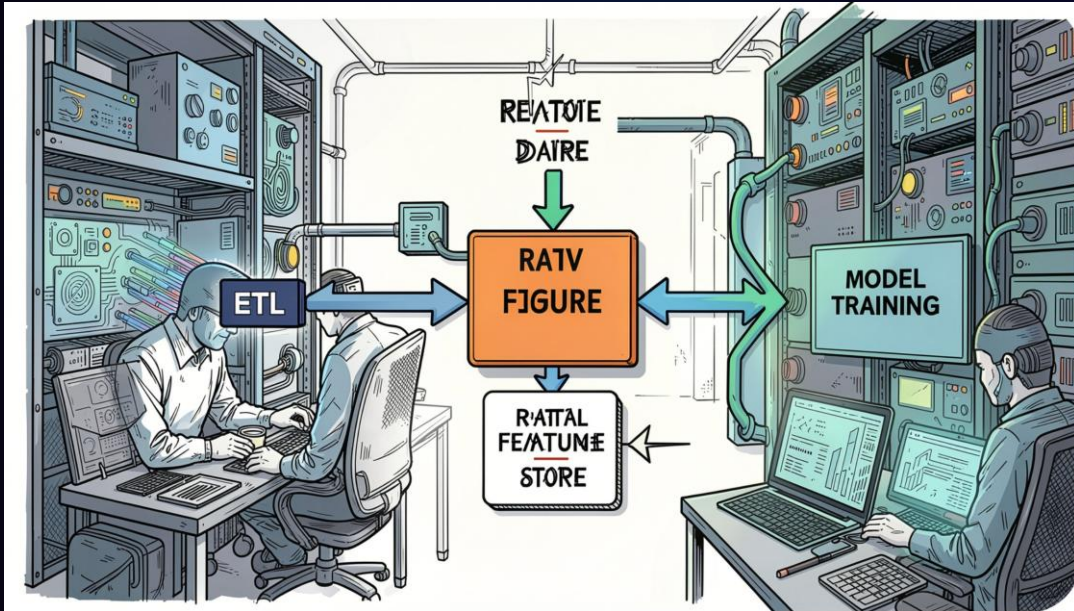
Experiments

ML models, NLP
pipeline,
visualizations

Results & Next Steps

Findings, limitations, roadmap

Project Objectives



Primary Goals

- Build reproducible ML pipelines for classification & regression
- Implement NLP preprocessing (TF-IDF) for sentiment tasks
- Compare classical ML and neural models; produce clear visual evaluations

Focus: clarity, reproducibility, and academic rigor.

Datasets & Features



Primary Tabular Dataset

Structured features: numerical & categorical variables; used for regression/classification tasks.



Text Corpus (NLP)

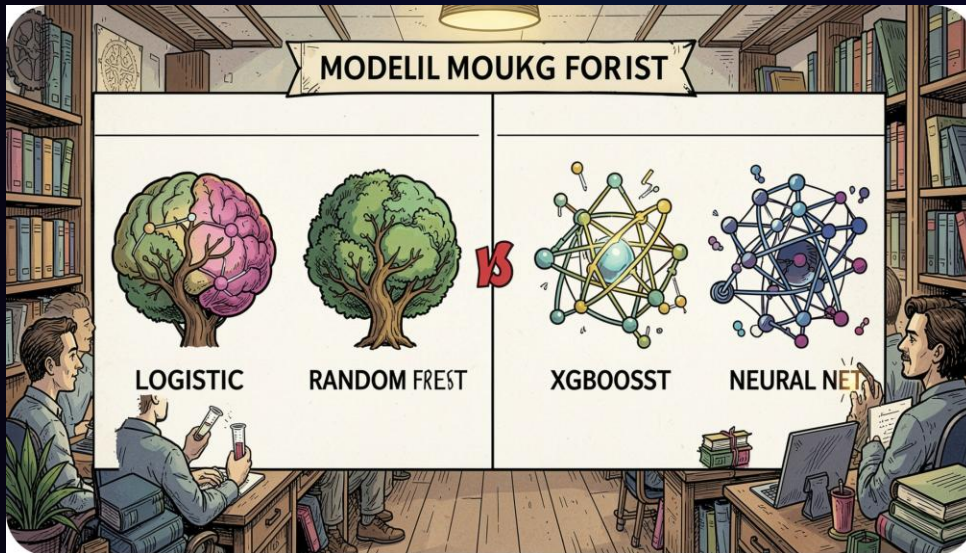
Short sentiment labels; high-cardinality and sparse labels noted — impacted NLP performance.



Engineered Features

TF-IDF vectors, numeric scaling, categorical encoding, basic embeddings for experiments.

Modeling Approach



01

Baseline Models

Logistic Regression, Random Forest — quick benchmarks.

02

Advanced Models

XGBoost and feedforward Neural Networks for improved fit.

03

NLP Pipeline

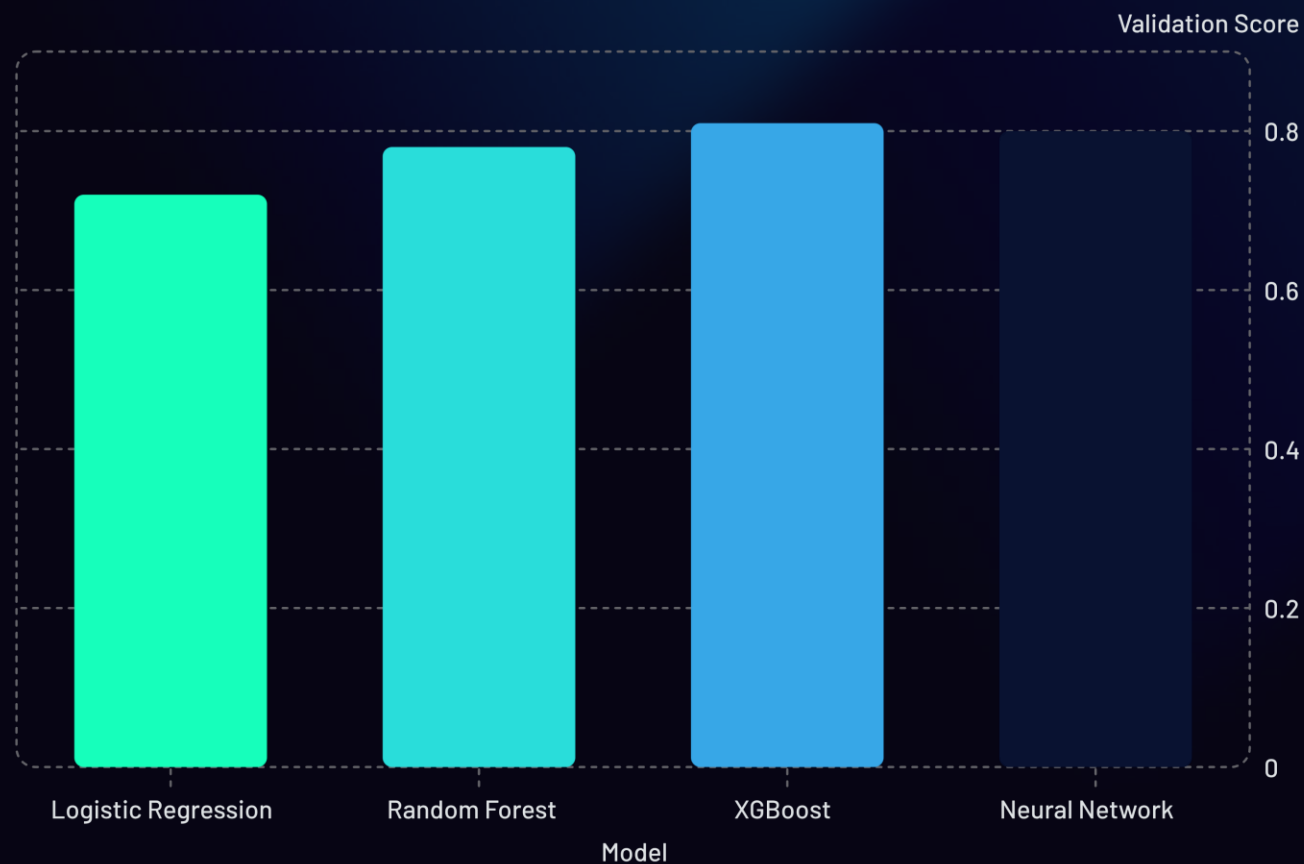
TF-IDF → classifier; noted label sparsity/high-cardinality constraints.

04

Evaluation

Cross-validation, confusion matrices, regression scatter plots.

Model Comparison (Validation)



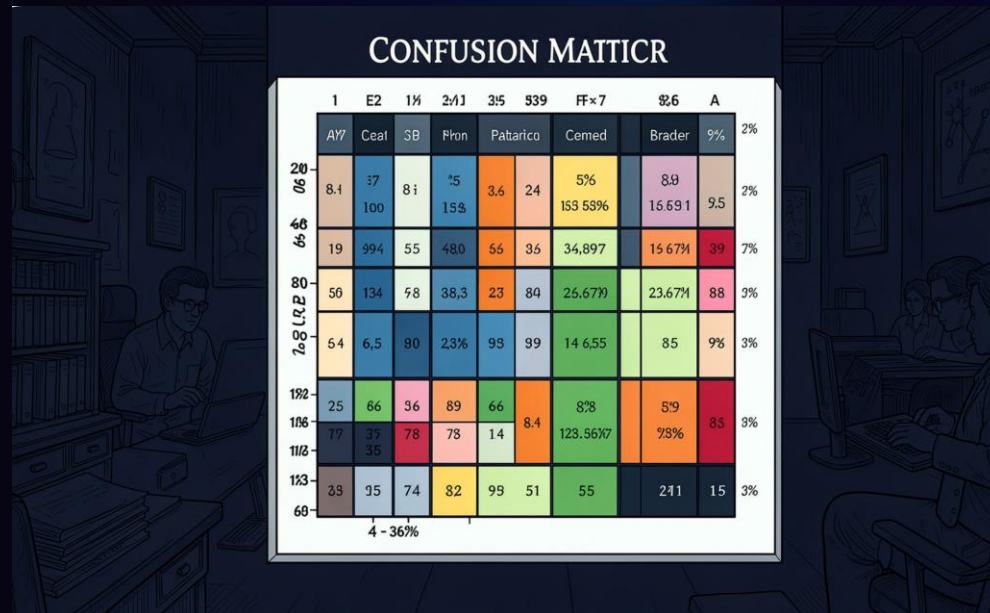
Notes: Scores are validation metrics from cross-validation. XGBoost leads by a small margin; NN close behind.

❏ Experimental results only — do not generalize beyond current data splits.

Classification Evaluation

Key Metrics

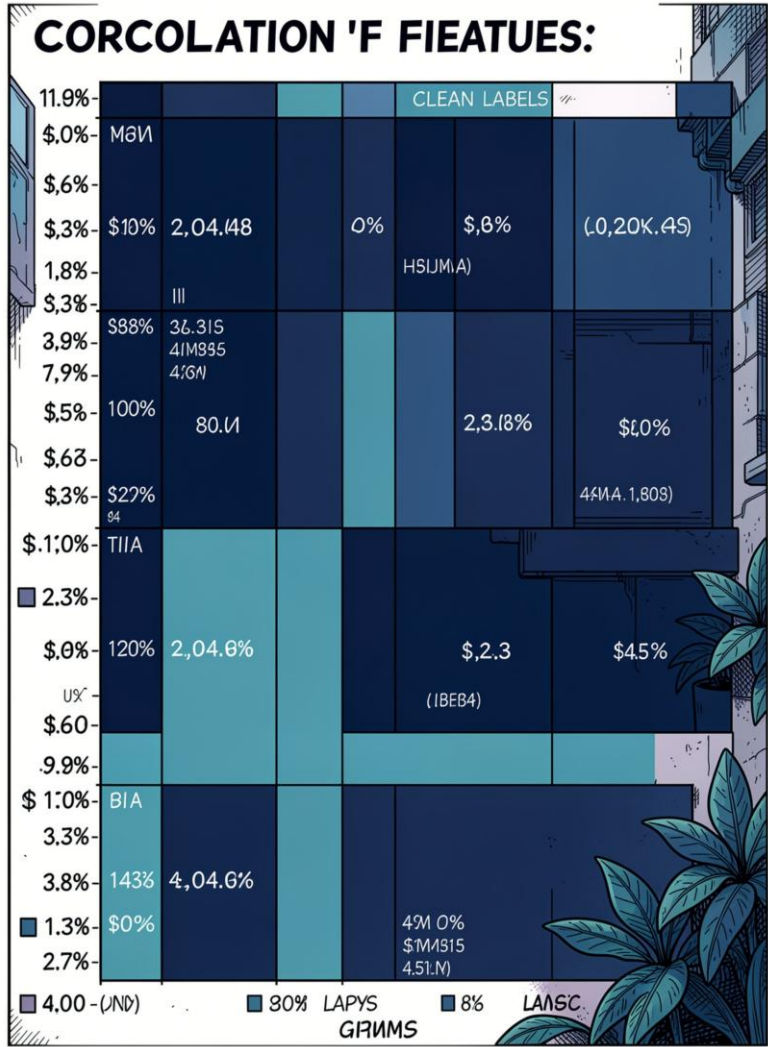
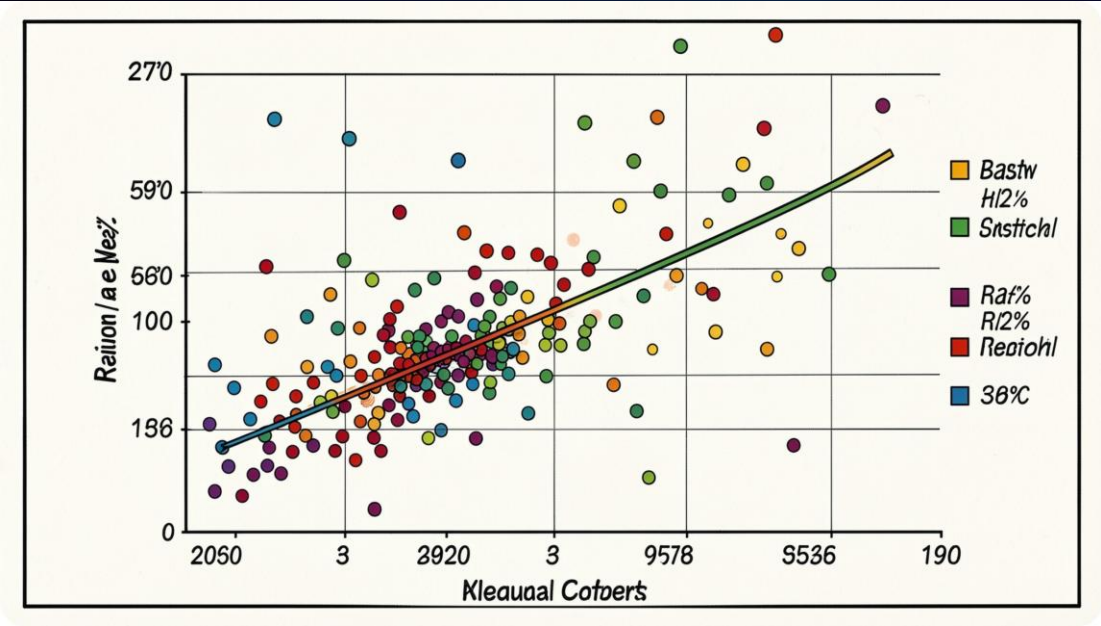
- Precision, Recall, F1 reported per class (detailed in viva)
- Confusion matrix used to identify class imbalance errors



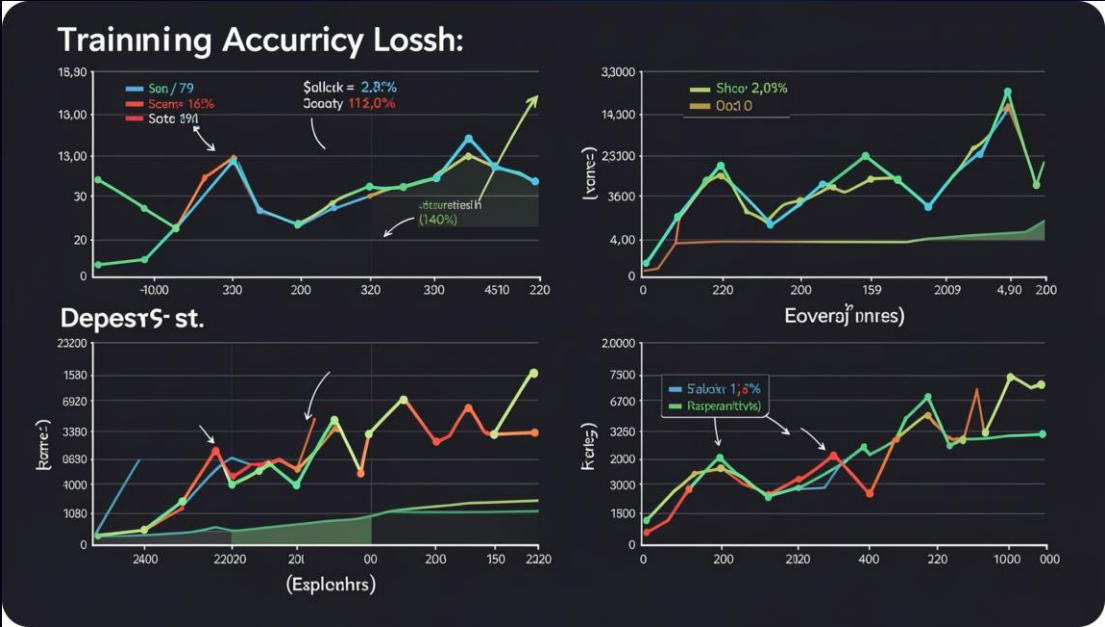
- ❗ NLP sentiment labels are sparse/high-cardinality — impacted classifier confidence.

Recommendation: consolidate rare labels or collect more labeled samples for robust NLP performance.

Regression & Correlation Analysis



Training Dynamics & Architecture



Conclusions, Limitations & Next Steps

Conclusions

XGBoost performed best on validation; neural models comparable. Pipelines are reproducible and modular.

Limitations

NLP performance limited by sparse / high-cardinality sentiment labels. Results are experiment-specific.

Next Steps

- Collect/merge labels to reduce sparsity
- Hyperparameter tuning and ensembling
- Deploy reproducible pipeline notebooks (Python / Jupyter)